ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 2 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable sketches.

SECTION A

QUESTION 1

1.1 1.1.1 C ✓ 1.1.2 C ✓ 1.1.3 A ✓ 1.1.4 D ✓ 1.1.5 B ✓ 1.1.6 D ✓ 1.1.7 B ✓ 1.1.8 C ✓ 1.1.9 A ✓ 1.1.10 B ✓ (10)

1.2 1.2.1 False ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 H ✓ 1.3.3 F ✓ 1.3.4 D ✓ 1.3.5 A ✓ 1.3.6 J ✓ 1.3.7 B ✓ 1.3.8 G ✓ 1.3.9 E ✓ 1.3.10 I ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Lines of force go straight across from the N pole to the S pole ✓, curving outwards near the edges ✓, with arrows from N to S ✓ (3)

2.2 Any THREE ✓✓✓: increase the current; increase the number of turns; use a soft-iron core; wind the turns closer together (3)

2.3 The small Arduino current flows through the relay coil ✓, which becomes an electromagnet ✓ and pulls the armature / moves the contacts ✓, closing a separate circuit that carries the large motor current (the two circuits are isolated) ✓. (4)

2.4 The emf induced in a conductor is proportional to the rate at which it cuts (the rate of change of) magnetic flux ✓✓. Factors ✓✓✓: strength of the magnetic field; speed of movement (rate of change); number of turns on the coil; (length of conductor in the field) (5)

[15]

QUESTION 3

3.1.1 Step-down ✓ (1)

3.1.2 $V_s \div V_p = N_s \div N_p$ ✓ $\rightarrow V_s = 230 \times 60 \div 1\,150$ ✓ $= 12$ V ✓ (3)

3.1.3 $V_p I_p = V_s I_s$ ✓ $\rightarrow I_p = 12 \times 2 \div 230$ ✓ $= 0{,}10$ A ✓ (3)

3.2 Block drawn in the correct order ✓ and its function ✓ for each of the five blocks: transformer ✓ – steps 230 V AC down to a low AC voltage ✓; rectifier ✓ – changes AC into pulsating DC ✓; filter / smoothing capacitor ✓ – smooths the pulsating DC and reduces ripple ✓; regulator (e.g. 7805) ✓ – keeps the output at a constant 5 V ✓; load / output ✓ – the robot circuit that uses the 5 V DC ✓ (10)

3.3 Half-wave: only the positive half-cycles, gaps between them ✓✓; full-wave: every half-cycle appears as a positive hump, no gaps (double the frequency) ✓✓ (4)

[21]

QUESTION 4

4.1 $V_L = \sqrt{3} \times V_{ph}$ ✓ $= \sqrt{3} \times 230 = 398{,}37$ V $(\approx 400$ V) ✓ (2)

4.2 Any TWO ✓✓: more power with less copper; three-phase motors are self-starting, smaller and smoother; constant power delivery; two voltages available (230 V and 400 V) (2)

4.3.1 $C_T = 10 + 20 + 20$ ✓ $= 50$ µF ✓ (2)

4.3.2 $1/C_T = 1/10 + 1/20 + 1/20$ ✓ $= 0{,}2$ ✓; $C_T = 5$ µF ✓✓ (4)

4.3.3 $Q = C \times V$ ✓ $= 50 \times 10^{-6} \times 12 = 6 \times 10^{-4}$ C $(600$ µC) ✓ (2)

[12]

QUESTION 5

5.1 Any FOUR (part ✓ + function ✓): yoke / frame – supports the motor and carries the magnetic flux; field poles / field winding or permanent magnets – produce the magnetic field; armature – rotating coils on which the force acts; commutator – reverses the current in the armature coils every half turn so it keeps turning; brushes – carry current to the commutator; bearings / shaft – allow rotation and deliver the mechanical output (8)

5.2 A motor converts electrical energy into mechanical (rotational) energy ✓; a generator converts mechanical energy into electrical energy ✓. (2)

5.3 Power supply ✓; CPU / processor (with memory) ✓; input module ✓; output module ✓ (accept programming device as one) (4)

5.4 Normally open START contact ✓ in series with normally closed STOP contact ✓ driving the motor output coil ✓; a normally open contact of the motor output in parallel with START (seal-in / latch) ✓✓; correct rails and labels ✓ (6)

5.5 Any TWO ✓✓: easy to change the program instead of rewiring; less wiring and smaller panels; fault-finding with the programming software; timers and counters built in; reliable in harsh environments; can communicate with networks / SCADA (2)

[22]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE